PREMIUM

Full-Length Mock Test

Mock 01

Production and Industrial Engineering

GATE PI

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | **Moderate 33%** | **Hard 47%**

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Manufacturing Processes — Casting]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.2 [1 Mark] [EASY] [Industrial Engineering — Inventory Control]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.3 [1 Mark] [EASY] [Metrology — Comparators]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.4 [1 Mark] [EASY] [Thermodynamics — Entropy]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.5 [1 Mark] [EASY] [Manufacturing Processes — Forming]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.6 [1 Mark] [EASY] [Industrial Engineering — Quality Control]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.7 [1 Mark] [EASY] [Metrology — CMM]

In sand casting, pattern is:

A) Same as casting

B) Larger than casting

C) Smaller than casting

D) Double the size

Q.8 [1 Mark] [EASY] [Thermodynamics — Refrigeration]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.9 [1 Mark] [EASY] [Manufacturing Processes — Forming]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.10 [1 Mark] [EASY] [Industrial Engineering — ERP]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.11 [1 Mark] [EASY] [Metrology — CMM]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.12 [1 Mark] [EASY] [Thermodynamics — Refrigeration]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.13 [2 Marks] [MODERATE] [Manufacturing Processes — Welding]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.14 [2 Marks] [MODERATE] [Industrial Engineering — Inventory Control]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula

B) Taylor's formula

C) Hooke's law

D) Coulomb's law

Q.15 [2 Marks] [MODERATE] [Metrology — CMM]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.16 [2 Marks] [MODERATE] [Thermodynamics — Entropy]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.17 [2 Marks] [MODERATE] [Manufacturing Processes — Casting]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.18 [2 Marks] [MODERATE] [Industrial Engineering — ERP]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.19 [2 Marks] [MODERATE] [Metrology — Measurement]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.20 [2 Marks] [MODERATE] [Thermodynamics — Refrigeration]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.21 [2 Marks] [MODERATE] [Manufacturing Processes — Welding]

Gear module m =:

A) D/T

B) T/D

C) D/T in mm

D) T/D in mm

Q.22 [2 Marks] [MODERATE] [Industrial Engineering — Inventory Control]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.23 [2 Marks] [MODERATE] [Metrology — Comparators]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.24 [2 Marks] [MODERATE] [Thermodynamics — Power Cycles]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.25 [2 Marks] [MODERATE] [Manufacturing Processes — Casting]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.26 [2 Marks] [MODERATE] [Industrial Engineering — ERP]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.27 [2 Marks] [MODERATE] [Metrology — CMM]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.28 [2 Marks] [MODERATE] [Thermodynamics — Power Cycles]

In sand casting, pattern is:

A) Same as casting

B) Larger than casting

C) Smaller than casting

D) Double the size

Q.29 [2 Marks] [MODERATE] [Manufacturing Processes — Casting]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.30 [2 Marks] [MODERATE] [Industrial Engineering — ERP]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.31 [2 Marks] [MODERATE] [Metrology — Limits & Fits]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.32 [2 Marks] [MODERATE] [Thermodynamics — Psychrometrics]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula
B) Taylor's formula
C) Hooke's law
D) Coulomb's law

Q.33 [2 Marks] [HARD] [Manufacturing Processes — Metal Cutting]

Gear module m =:

A) D/T
B) T/D
C) D/T in mm
D) T/D in mm

Q.34 [2 Marks] [HARD] [Industrial Engineering — ERP]

In sand casting, pattern is:

A) Same as casting
B) Larger than casting
C) Smaller than casting
D) Double the size

Q.35 [2 Marks] [HARD] [Metrology — Measurement]

Shear angle in orthogonal cutting is given by:

A) Merchant's formula

B) Taylor's formula

C) Hooke's law

D) Coulomb's law

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Thermodynamics — Laws]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Manufacturing Processes — Welding]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Industrial Engineering — Quality Control]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Metrology — Surface Finish]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Thermodynamics — Entropy]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Metrology — Comparators]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: ____________________

Q.52 [2 Marks] [HARD] [Thermodynamics — Psychrometrics]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: ____________________

Q.53 [2 Marks] [HARD] [Manufacturing Processes — Casting]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: ____________________

Q.54 [2 Marks] [HARD] [Industrial Engineering — Inventory Control]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: ____________________

Q.55 [2 Marks] [HARD] [Metrology — Measurement]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: ____________________

Q.56 [2 Marks] [HARD] [Thermodynamics — Refrigeration]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: ____________________

Q.57 [2 Marks] [HARD] [Manufacturing Processes — Machining]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: ____________________

Q.58 [2 Marks] [HARD] [Industrial Engineering — Quality Control]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: ______________________

Q.59 [2 Marks] [HARD] [Metrology — CMM]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: ______________________

Q.60 [2 Marks] [HARD] [Thermodynamics — Refrigeration]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: ______________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	A	**Q.3:**	C
Q.4:	B	**Q.5:**	A	**Q.6:**	C
Q.7:	B	**Q.8:**	A	**Q.9:**	C
Q.10:	B	**Q.11:**	A	**Q.12:**	C
Q.13:	B	**Q.14:**	A	**Q.15:**	C
Q.16:	B	**Q.17:**	A	**Q.18:**	C
Q.19:	B	**Q.20:**	A	**Q.21:**	C
Q.22:	B	**Q.23:**	A	**Q.24:**	C
Q.25:	B	**Q.26:**	A	**Q.27:**	C
Q.28:	B	**Q.29:**	A	**Q.30:**	C
Q.31:	B	**Q.32:**	A	**Q.33:**	C
Q.34:	B	**Q.35:**	A	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	100
Q.52:	447	**Q.53:**	100	**Q.54:**	447
Q.55:	100	**Q.56:**	447	**Q.57:**	100
Q.58:	447	**Q.59:**	100	**Q.60:**	447

Detailed Solutions

Question 1
[Manufacturing Processes — Casting]

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 2
[Industrial Engineering — Inventory Control]

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 3
[Metrology — Comparators]

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 4
[Thermodynamics — Entropy]

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 5
[Manufacturing Processes — Forming]

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 6
[Industrial Engineering — Quality Control]

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 7
[Metrology — CMM]

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 8
[Thermodynamics — Refrigeration]

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 9
[Manufacturing Processes — Forming]

Gear module m =:

Answer:

(C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 10 *[Industrial Engineering — ERP]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 11 *[Metrology — CMM]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 12 *[Thermodynamics — Refrigeration]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 13 *[Manufacturing Processes — Welding]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 14 *[Industrial Engineering — Inventory Control]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 15 *[Metrology — CMM]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 16 *[Thermodynamics — Entropy]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 17 *[Manufacturing Processes — Casting]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 18 *[Industrial Engineering — ERP]*

Gear module m =:

Answer:

(C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 19 *[Metrology — Measurement]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 20 *[Thermodynamics — Refrigeration]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 21 *[Manufacturing Processes — Welding]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 22 *[Industrial Engineering — Inventory Control]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 23 *[Metrology — Comparators]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 24 *[Thermodynamics — Power Cycles]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 25 *[Manufacturing Processes — Casting]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 26 *[Industrial Engineering — ERP]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[² ±/2 - ;"ó"à

Question 27 *[Metrology — CMM]*

Gear module m =:

Answer:

(C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 28 *[Thermodynamics — Power Cycles]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 29 *[Manufacturing Processes — Casting]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[^{2} ±/2 - ;"ó"à

Question 30 *[Industrial Engineering — ERP]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 31 *[Metrology — Limits & Fits]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 32 *[Thermodynamics — Psychrometrics]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[^{2} ±/2 - ;"ó"à

Question 33 *[Manufacturing Processes — Metal Cutting]*

Gear module m =:

Answer: (C)

m = D/T (mm). Module is pitch diameter in mm divided by teeth.

Question 34 *[Industrial Engineering — ERP]*

In sand casting, pattern is:

Answer: (B)

Pattern is oversized to account for metal shrinkage (shrink allowance).

Question 35 *[Metrology — Measurement]*

Shear angle in orthogonal cutting is given by:

Answer: (A)

Merchant's theory: <b Ò C[^{2} ±/2 - ;"ó"à

Question 36 *[Thermodynamics — Laws]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Manufacturing Processes — Welding]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 38 *[Industrial Engineering — Quality Control]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 39 *[Metrology — Surface Finish]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 40 *[Thermodynamics — Entropy]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 41 *[Manufacturing Processes — Machining]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 42 *[Industrial Engineering — ERP]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 43 *[Metrology — Surface Finish]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 44 *[Thermodynamics — Power Cycles]*
Which statements are correct?
Answer: (A, C)
Based on theoretical analysis of the topic.

Question 45 *[Manufacturing Processes — Forming]*
Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Industrial Engineering — Ergonomics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Metrology — Comparators]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Thermodynamics — Psychrometrics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Manufacturing Processes — Casting]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Industrial Engineering — Inventory Control]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Metrology — Comparators]*

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: (100)

D = m×T = 5×20 = 100mm.

Question 52 *[Thermodynamics — Psychrometrics]*

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: (447)

EOQ = " (2DS/H) = " (2×10000×50/5) = " 200000 "H 447.

Question 53 *[Manufacturing Processes — Casting]*

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: (100)

D = m×T = 5×20 = 100mm.

Question 54 *[Industrial Engineering — Inventory Control]*

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer:

EOQ = " (2DS/H) = " (2×10000×50/5) = " 200000 "H 447.

Question 55

[Metrology — Measurement]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: (100)

D = m×T = 5×20 = 100mm.

Question 56

[Thermodynamics — Refrigeration]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: (447)

EOQ = " (2DS/H) = " (2×10000×50/5) = " 200000 "H 447.

Question 57

[Manufacturing Processes — Machining]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: (100)

D = m×T = 5×20 = 100mm.

Question 58

[Industrial Engineering — Quality Control]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: (447)

EOQ = " (2DS/H) = " (2×10000×50/5) = " 200000 "H 447.

Question 59

[Metrology — CMM]

Gear module = 5, teeth = 20. Pitch diameter (mm)?

Answer: (100)

D = m×T = 5×20 = 100mm.

Question 60

[Thermodynamics — Refrigeration]

EOQ: D=10000/year, H=5/unit/year, S=50/order. EOQ = ?

Answer: (447)

EOQ = " (2DS/H) = " (2×10000×50/5) = " 200000 "H 447.
